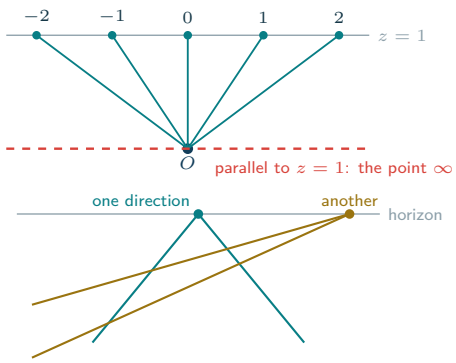
PROJECTIVE GEOMETRY LADDER 1 OF 9

The point at infinity

Parallel lines never meet in the plane, and adding one point at infinity for each direction removes the exception.

1 / THE RULE AND THE PICTURE

Add one point for every direction, and every two lines meet in exactly one point



A point of the projective line is a line through the origin, and each such line meets the line $z = 1$ at one number x, except the one line parallel to it, which supplies the single new point ∞:

$$\mathbb{P}^1(F) = F \cup \{\infty\}$$

In the plane the same idea adds a whole line of new points, one per direction, as railway lines show when they appear to meet on the horizon.

2 / THREE WAYS TO MEET A POINT AT INFINITY

As a limit, as a direction and as a line through the origin

Walk along the line $y = 2x + 1$ and the point $(t, 2t + 1)$ runs off to infinity, but in the coordinates $(t : 2t + 1 : 1) = (1 : 2 + \frac{1}{t} : \frac{1}{t})$ it settles down to $(1 : 2 : 0)$, the point at infinity of that direction.
Every line with slope 2 shares that limit, which is why parallel lines meet there.
The same point is the direction $(1, 2)$ and the line through the origin of three-dimensional space with direction $(1, 2, 0)$.

3 / THE LINE AT INFINITY

One point for each slope, so nothing is left over

The new points $(a : b : 0)$ form a line, because they all satisfy $Z = 0$, and two lines of the plane meet at infinity exactly when they have the same slope.
Two distinct lines now always share exactly one point: the lines $y = 2x + 1$ and $y = 2x + 5$ meet at $(1 : 2 : 0)$, and the lines $y = 2x + 1$ and $y = -x$ meet at the ordinary point $(-\frac{1}{3}, \frac{1}{3})$.

4 / THE PROJECTIVE LINE ON ITS OWN

A circle over the reals, a sphere over the complex numbers

Over the real numbers the line $\mathbb{P}^1(\mathbb{R})$ is a circle, since turning a direction through $180°$ gives the same line and the two ends of the number line join at ∞.
Over the complex numbers it is the Riemann sphere, and over a field with p elements it has $p + 1$ points, the p numbers and ∞.

5 / TRY THESE

Convert between the two descriptions

(a) The triple $(4 : 6 : 2)$ is the point $(2, 3)$, found by dividing by the last entry. **(b)** The triple $(3 : 6 : 0)$ has last entry 0, so it is the direction of slope 2, the same point as $(1 : 2 : 0)$. **(c)**
The triple $(0 : 0 : 0)$ is not a point, since it describes no line through the origin.

ASIDE / WHY BOTHER

Every statement that used to need the exception "unless the lines are parallel" now holds without it, which is what makes the counting theorems on the later sheets possible.

LINKS TO OTHER SHEETS

The next sheet turns the idea into coordinates; the algebra ladder shows the same laws of arithmetic that this geometry rests on.

Convention. F is a field; $\mathbb{P}^1(F)$ is the projective line over F; the point $(a : b : c)$ is the set of non-zero multiples of (a, b, c).